Strategy 6

Opponent-Adjusted Matchup Modeling

Mathematical documentation and system framework

Applications: NBA, NFL, NCAA Division I FBS Football (NCAAF).

Objective: Estimate player-level predictive distributions by accounting for defensive quality, opponent tactics, positional matchups, personnel, and game environment.

1. Executive summary

Opponent-adjusted modeling modifies a player's neutral baseline to reflect the upcoming defense. A raw defensive ranking is insufficient: opponent quality, play volume, injuries, schedule composition, and schemes all affect observed production. Prefer adjustments to opportunity and efficiency components over a blanket adjustment to the final statistic.

$$\mu_{\text{adjusted}} = \mu_{\text{baseline}} M_{\text{opponent}}$$

Term	Meaning
Baseline	Expected production against neutral opposition
Opponent multiplier	Matchup-specific relative change, ideally estimated out of sample
Adjusted expectation	Conditional projected statistical mean

Illustration: A 65-yard receiving baseline multiplied by 1.12 yields 72.8 expected yards. The 1.12 value must be estimated, not assigned from a rank alone.

2. Core mathematical framework

2.1 Neutral player baseline

$$\mu_{p,s}^{(0)} = \sum_{i=1}^{N} w_i X_{p,s,i}, \qquad \sum_i w_i = 1$$

Here p denotes player, s the prop statistic, X a prior performance, and w a normalized historical weight. Historical games can first be normalized for prior opponents; otherwise an upcoming defensive adjustment may double-count old matchup effects.

2.2 Opportunity-efficiency decomposition

$$E[X] = E[O]E[\eta] + \text{Cov}(O, \eta)$$

O is opportunity (minutes, routes, attempts, touches) and eta is production per opportunity. The covariance is relevant when opportunity and efficiency move together. For NFL receiving yards, a useful structural approximation is routes times targets per route times catches per target times yards per catch.

2.3 Simple opponent factor

$$M_{o,s} = D_{o,s} / D_{\text{league},s}$$

Use this ratio only when the numerator and denominator share the same units and opportunity basis. For a hypothetical opponent allowing 168 wide-receiver yards versus a 150-yard league average, M = 1.12. Higher allowed production corresponds to an easier matchup, all else equal.

Factor	Illustrative interpretation
0.85	15% below neutral production
1.00	Neutral
1.10	10% above neutral production
1.20	20% above neutral production

3. Opponent-strength and schedule adjustments

3.1 Actual versus expected

$$R_{o,g} = A_{o,g} / E_{o,g}$$

$$M_o = \sum_g A_{o,g} / \sum_g E_{o,g}$$

A is production actually allowed by defense o in game g; E is expected production based on the prior opponent's neutral offense and game opportunity. Normalize to per-play or per-possession rates where possible to avoid confusing tempo with defensive quality.

Game	Expected passing yards	Actual allowed	Actual / expected
1	250	275	1.100
2	210	231	1.100
3	300	270	0.900

Aggregate factor = 776 / 760 = 1.021053, about 2.1% above expectation.

3.2 Recency weighting

$$w_g = e^{-\lambda a_g} / \sum_j e^{-\lambda a_j}$$

$$M_o = \sum_g w_g A_{o,g} / \sum_g w_g E_{o,g}$$

Age a is measured in a consistent unit; lambda sets decay. Recent injury or scheme shifts may justify shorter windows. Weights require walk-forward estimation, not information from later games.

3.3 Small-sample shrinkage

$$\widetilde{M}_o = \alpha_o M_o + (1 - \alpha_o), \quad \alpha_o = n_{\text{eff}}/n_{\text{eff}} + k$$

$$n_{\text{eff}} = (\sum_g w_g)^2 / \sum_g w_g^2$$

k is prior strength; this pulls sparse estimates toward a neutral factor of 1. For raw M = 1.20, effective sample size 4, and k = 6, alpha = 0.40 and shrunk M = 1.08.

4. NBA matchup modeling

4.1 Points: attempts and conversion

$$E[PTS] = 2E[2PA]p_{2P} + 3E[3PA]p_{3P} + E[FTA]p_{FT}$$

Estimate opponent effects separately on two- and three-point attempts, field-goal probabilities, and free throws. Relevant inputs include rim protection, three-point attempt share allowed, scheme, shot-quality allowed, and pace. Keep adjusted probabilities inside [0,1]; logistic links are suitable.

Example: Expected 3PA = 7; 3P probability = .38. If matchup raises attempt volume 12% and decreases success probability 5%, adjusted made threes = 7(1.12)(.38)(.95) = 2.83024, versus 2.66 neutral.

4.2 Rebounds

$$E[REB_p] = E[R_{\text{available}}]P(p \text{ secures rebound})$$

Available rebounds depend on missed field goals and reboundable free throws. Rebound share depends on lineup, positioning, missed-shot locations, opponent offensive/defensive rebounding rates, and playing time. A baseline of 9.2 rebounds with 1.08 opportunity and 1.04 share factors becomes 10.33344; ensure factors are estimated conditionally to avoid overlap.

4.3 Assists

$$E[AST] = E[\text{potential assists}]P(\text{conversion})$$

Defensive help, double teams, shot profile, and teammate shooting determine both passing chances and conversion. A trap-heavy defense might raise passing opportunity without necessarily raising realized assists.

4.4 Steals, blocks, turnovers

Statistic	Key opposing inputs
Steals	Opponent turnover rate, ball-handling, live-ball turnovers
Blocks	Rim attempts faced, opponent shot profile, shot-block susceptibility
Turnovers	Pressure, traps, steal rate, player ball-handling exposure
Three-pointers	Attempt volume allowed, contests, perimeter scheme

$$X \sim \text{Poisson}(\lambda), \quad P(X \geq 2) = 1 - e^{-\lambda}(1 + \lambda)$$

Poisson is a starting point for low-frequency events; test overdispersion and excess zeros before adopting it.

5. NFL matchup modeling

5.1 Quarterback passing yards

$$E[PY]_{\text{adj}} = E[PA]_0 M_{\text{volume}}\ E[YPA]_0 M_{\text{efficiency}}$$

Use EPA allowed per dropback, success rate, pressure, blitz, explosive passes, and opponent pass-rate-over-expected. Pressure and blitz effects should interact with quarterback-specific response rather than assuming a universal negative effect.

5.2 Wide receiver receiving yards

$$E[RY] = E[R]P(T|R)P(C|T)E[Y|C]$$

Account for outside/slot alignment, man-zone mix, cornerback/safety assignments, route types, explosive plays, and pressure-adjusted throw depth. The opponent modifies routes, target rate, catch probability, and yards per catch separately.

Component	Neutral	Factor
Routes	35	1.00
Targets per route	0.24	1.08
Catch rate	0.68	0.97
Yards per catch	12.5	1.10

Adjusted receiving yards = 35 x .24 x 1.08 x .68 x .97 x 12.5 x 1.10 = 74.66004 yards. Cap probabilities appropriately.

5.3 Running back rushing yards

$$E[RY]_{\text{adj}} = E[RA]_0 M_{\text{volume}}\ E[YPC]_0 M_{\text{efficiency}}$$

Prioritize rush EPA/success rate allowed, run stuffs, explosive rates, defensive personnel, yards before contact, and expected game script. Raw yards allowed per game blends opponent attempts with defensive efficiency.

5.4 Tight ends and receiving backs

Add route location, linebacker and safety coverage, middle-of-field targets, short-pass efficiency, and allowed running-back receptions. Position splits with limited samples warrant stronger shrinkage.

6. NCAAF matchup modeling

College football requires stronger schedule normalization because opponent quality, schemes, conference strength, roster composition, and blowout likelihood vary considerably.

6.1 Hierarchical defensive strength

$$D_o = \mu_D + \gamma_{\text{conference}(o)} + u_o$$

The national baseline, conference effect, and team-specific effect share information across related observations. Estimate conference effects from a connected cross-conference schedule, not subjective fixed ratings.

6.2 Passing and rushing

$$E[PY] = E[DB]P(PA|DB)E[YPA]_{\text{adj}}$$

$$E[RY] = E[RA]_{\text{adj}} E[YPC]_{\text{adj}}$$

For quarterbacks, use opponent-adjusted passing EPA, explosive pass allowed, pressure, and offensive system. For runners, incorporate offensive-line versus defensive-front interactions, mobile-quarterback usage, and expected substitutions.

NCAAF factor	Required treatment
Uneven schedules	Normalize prior opponents
Conference gaps	Hierarchical estimates
Small samples	Stronger shrinkage
Blowouts	Scenario-adjusted playing time
Roster turnover	Downweight obsolete history

7. Multifeature opponent multiplier

$$M_{p,o,s} = \exp\left(\sum_{j=1}^{K} \beta_{s,j} z_{p,o,j}\right)$$

$$z_j = (x_j - \mu_j)/\sigma_j$$

Standardized inputs may represent defensive efficiency, pace, scheme, positional assignments, and relevant interactions. Fit coefficients with regularization and out-of-sample validation. Closely related metrics can double-count a single defensive effect.

Hypothetical NBA points feature	z	Coefficient
Defensive efficiency weakness	0.80	0.050
Opponent pace	0.50	0.030
Rim protection strength	1.20	-0.025

Multiplier = exp(.050 x .80 + .030 x .50 - .025 x 1.20) = exp(.025) = 1.025315. A neutral 25-point projection becomes approximately 25.6329 points. Coefficients here are illustrative only.

8. Injuries and defensive lineup scenarios

$$E[X] = \sum_{k=1}^{K} P(L_k) E[X|L_k]$$

Personnel changes should be represented relative to the lineup already implicit in the defensive baseline. For an uncertain NBA center with 60% playing probability, blend active and inactive scenario projections 0.60 / 0.40. Avoid classifying unconfirmed status as certain.

9. Predictive distributions and prop probabilities

9.1 Normal approximation

$$X \sim N(\mu_{\text{adj}}, \sigma^2_{\text{adj}})$$

$$P(X > L) = 1 - \Phi\left(L - \mu_{\text{adj}}/\sigma_{\text{adj}}\right)$$

Use only when empirically justified. Integer statistics may require discrete distributions or continuity corrections.

9.2 Negative Binomial for overdispersion

$$E[X] = r(1-p)/p, \quad \mathrm{Var}(X) = r(1-p)/p^2$$

Can handle count outcomes with variance above the mean. For very rare events, investigate zero-inflation and other count models.

9.3 Monte Carlo

For each simulation: sample playing time/opportunity, defensive strength, lineup scenario and conditional efficiency; then generate the game statistic. Preserve dependencies rather than sampling every component independently.

$$\widehat{\mu} = 1/B \sum_{b=1}^{B} X^{(b)}$$

$$\widehat{P}(X > L) = 1/B \sum_{b=1}^{B} \mathrm{I}\{X^{(b)} > L\}$$

Treat pushes separately for integer prop lines.

10. Integrated modeling pipeline

Stage	Responsibility
Weighted history	Neutral player ability and recent form
Usage and opportunity	Expected minutes, snaps, routes, touches, attempts
Opponent matchup	Adjust opportunity and efficiency
Lineup scenarios	Capture uncertain personnel and assignments
Simulation	Produce a full predictive distribution
Market evaluation	Estimate line probabilities and calibration

Do not apply a second full pace factor if expected possessions already embed matchup tempo. The same caution applies to injuries, weather, and game script.

11. Proposed Python system architecture

Recommended modules: models/weighted_historical.py, usage_opportunity.py, opponent_adjustment.py, lineup_adjustment.py, simulation.py; features/defensive_efficiency.py, schedule_strength.py, positional_matchups.py, personnel.py; evaluation/backtesting.py, calibration.py, model_comparison.py; app/player_projection.py and matchup_analytics.py.

Input	Fields
Player history	Game stats, minutes/snaps, opportunities, available-at timestamps
Team defense	Per-play/per-possession metrics, positional splits, schemes
Prior opponents	Quality, historical expectations, actual allowed
Personnel	Likely starters, injuries, rotations, defensive assignments
Market	Timestamped prop lines and prices for comparison

Store prediction timestamps and source availability timestamps to prevent look-ahead bias. Outputs should include neutral projection, opponent multiplier, adjusted mean, dispersion, simulated outcomes, probabilities by prop line, and uncertainty diagnostics.

12. Backtesting and validation

$$MAE = 1/N \sum_{i=1}^{N} |X_i - \widehat{X}_i|$$

$$RMSE = \sqrt{1/N \sum_{i=1}^{N} (X_i - \widehat{X}_i)^2}$$

$$BS = 1/N \sum_{i=1}^{N} (p_i - y_i)^2$$

Compare weighted-history baseline; baseline plus usage/opportunity; and the same model plus opponent adjustments. Evaluate incremental performance out of sample by sport and prop category. For probabilistic forecasts, use calibration curves, Brier score, log loss, interval coverage and CRPS where suitable.

Walk-forward protocol

Train only on records available before each prediction time; predict the next slate; freeze forecasts; evaluate outcomes; advance the window. Inspect error by matchup multiplier bucket to diagnose over-aggressive favorable/unfavorable ratings.

13. Implementation roadmap

Phase	Scope	Priority
1	Schedule-adjusted defense	Essential
2	Recency and shrinkage	Essential
3	Opportunity / efficiency components	Essential
4	Position-specific matchups	High
5	Personnel scenarios	High
6	Tactical interactions	Medium
7	NCAAF hierarchical schedule model	High
8	Simulated distributions	High
9	Walk-forward calibration	Essential

14. Final mathematical summary

$$\mu_{p,o,s}^{\text{final}} = \mu_{p,s}^{(0)} \exp\left(\sum_j \beta_{s,j} z_{p,o,j}\right)$$

$$E[X_{p,o}] = E[O_{p,o}]E[\eta_{p,o}] + \text{Cov}(O_{p,o}, \eta_{p,o})$$

Key principle: estimate how the opponent changes the mechanisms creating a player statistic, not merely how far above or below average the defense ranks. Matchup factors are statistical estimates with uncertainty; their value must be demonstrated in chronological out-of-sample evaluation.